OData Service Creation and Administration

Public



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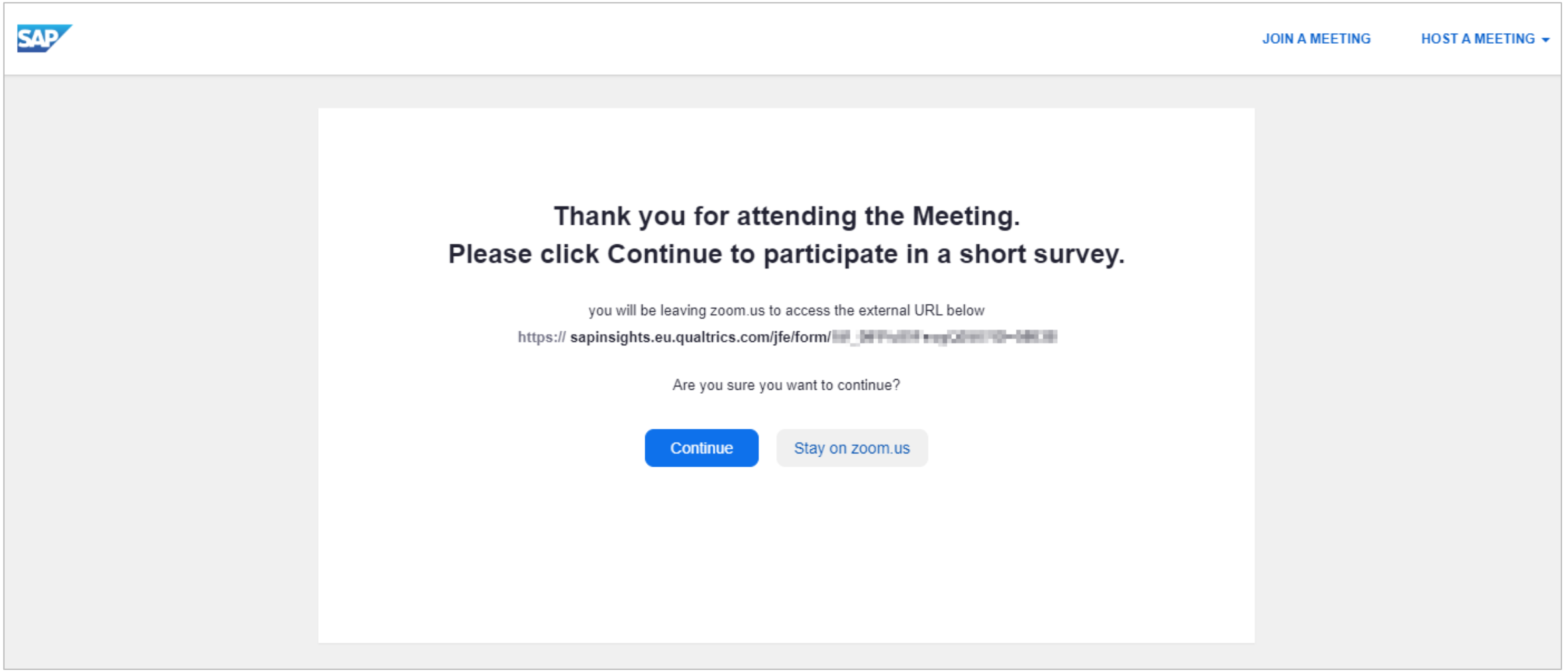


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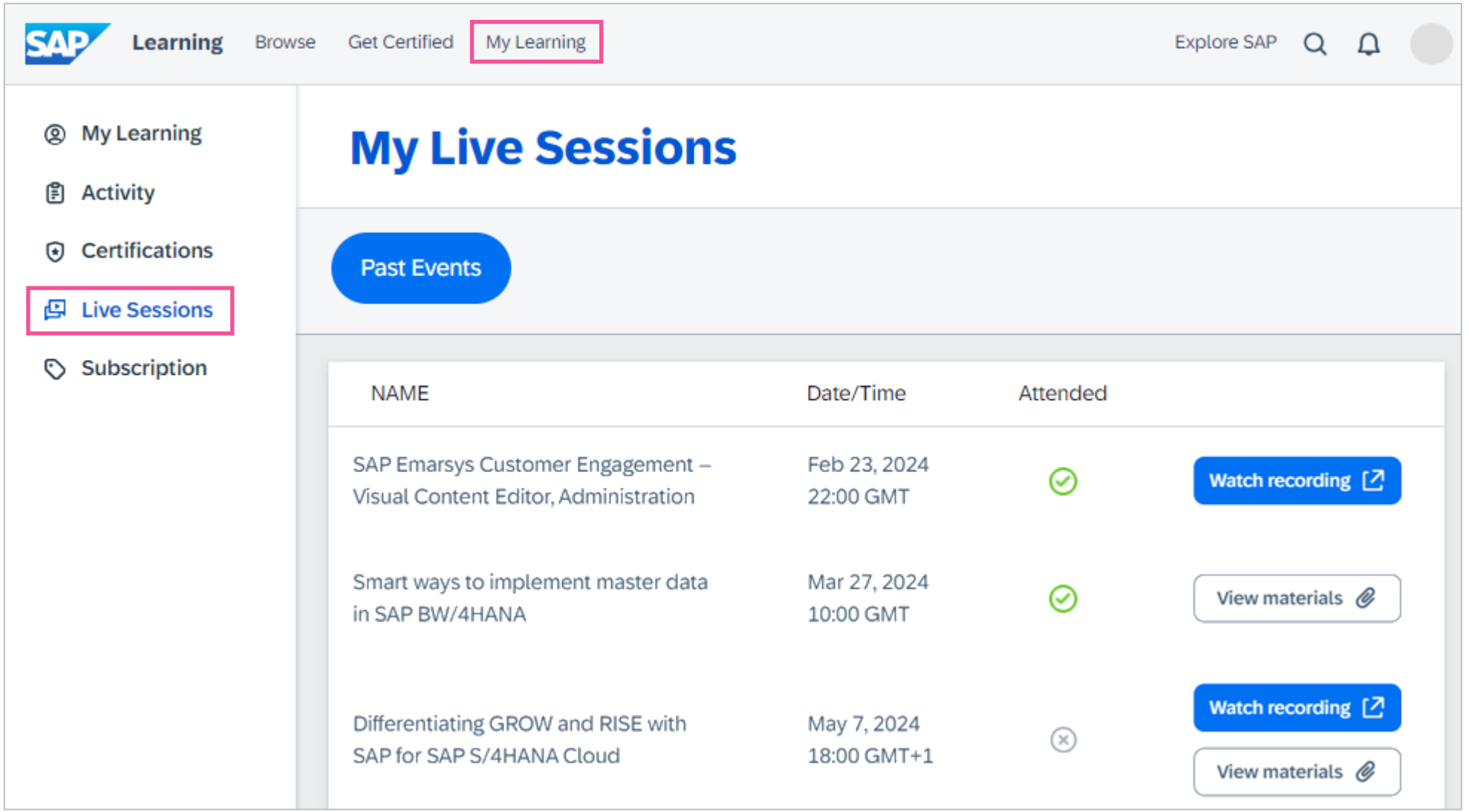


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Course Outline



Introduction to OData in SAP

- Historical Context and Evolution of OData
- Core Principles of OData in SAP



Generating OData V2 Services from CDS

- Introduction to Core Data Services
- Steps to Generate OData V2 Services



OData Service Definitions and Bindings

- Understanding OData V2 Service Definitions
- Binding OData V2 Services
- Differences between V2 and V4
- Binding OData V4 Services



Enhancing OData Services with Annotations

- Overview of OData Annotations



Practical Application and Best Practices

- Real-World Use Cases of OData in SAP
- Best Practices for OData Service Deployment

Introduction

This course offers a comprehensive overview of traditional and modern OData approaches in SAP. Gain foundational knowledge and hands-on experience with core OData principles, including historical context and evolution.

Learn to generate OData V2 services from Core Data Services (CDS) views and understand the definitions and bindings of OData services in both V2 and V4.

Explore essential annotations to enhance and customize OData services.

By the end of this course, you'll bridge the gap between traditional methodologies and modern practices, effectively utilizing OData within SAP systems.

Introduction to OData in SAP

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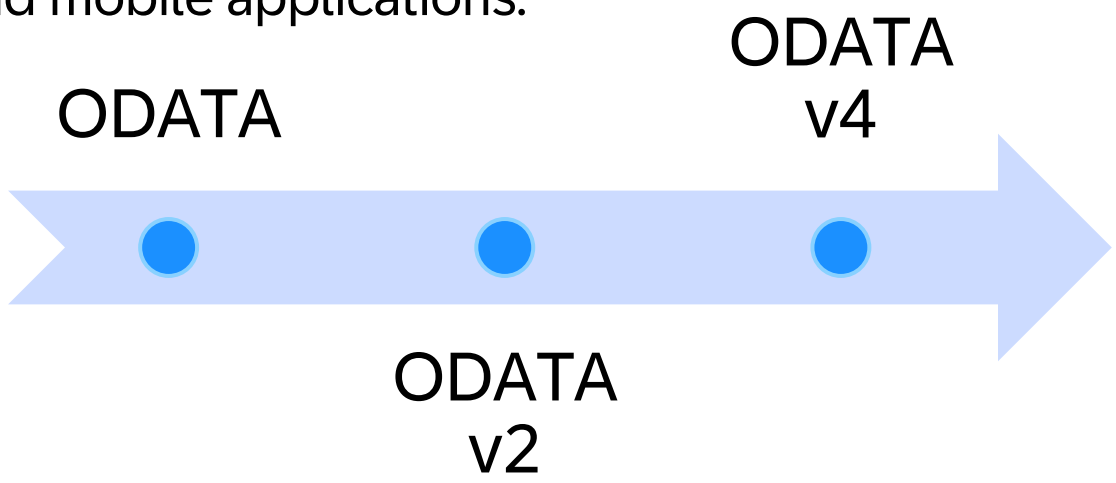


Historical Context and Evolution of OData

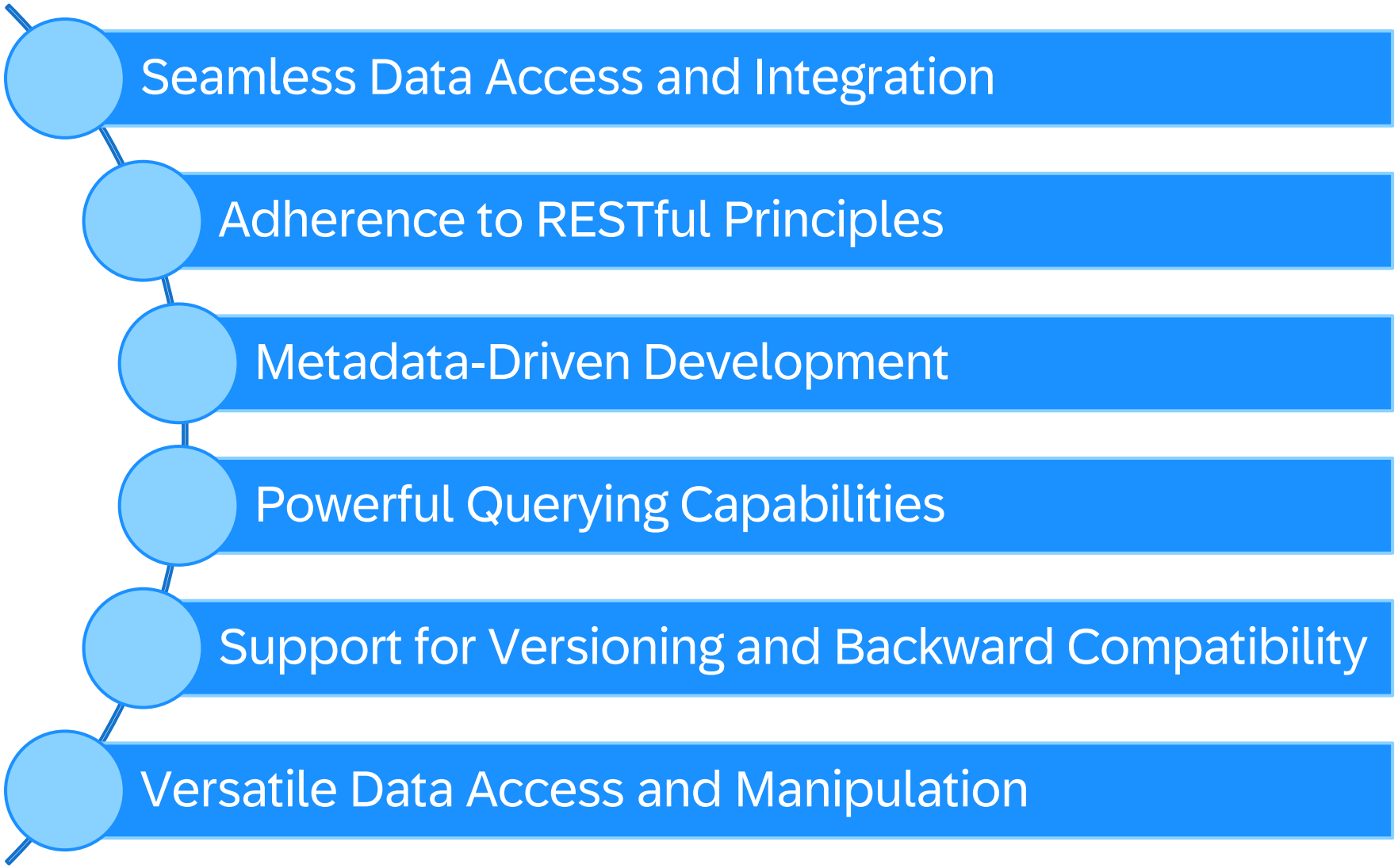
The Open Data Protocol (OData) addressed the need for a standardized protocol to query and update data across diverse systems.

Over the years, OData evolved through several versions, enhancing functionality to meet changing developer and enterprise demands. Early versions laid the foundational concepts, while later versions introduced support for complex data types, request batching, and robust query options. Widespread adoption, including integration with SAP systems, marked Version 2.

OData Version 4 brought significant improvements in performance, scalability, and flexibility, making it ideal for modern web and mobile applications.



Core Principles of OData in SAP



Generating OData V2 Services from CDS

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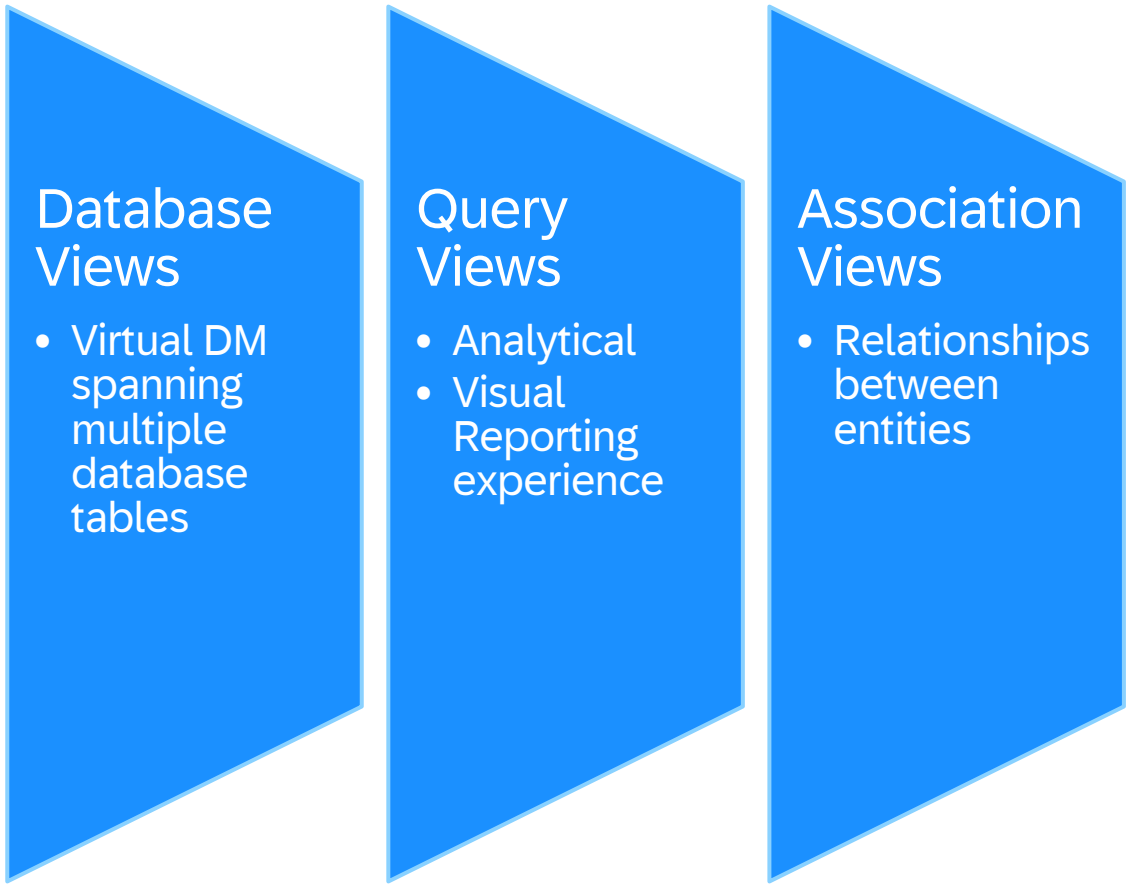


Introduction to Core Data Services (CDS)

Core Data Services (CDS) is a foundational technology within SAP that empowers developers to define and consume semantically rich data models in a unified manner.

It serves as the backbone for various SAP technologies, including OData service generation, ABAP programming, and SAP Fiori applications.

CDS adopts a declarative approach, leveraging a domain-specific language (DSL) that simplifies the development of intricate data structures and relationships.



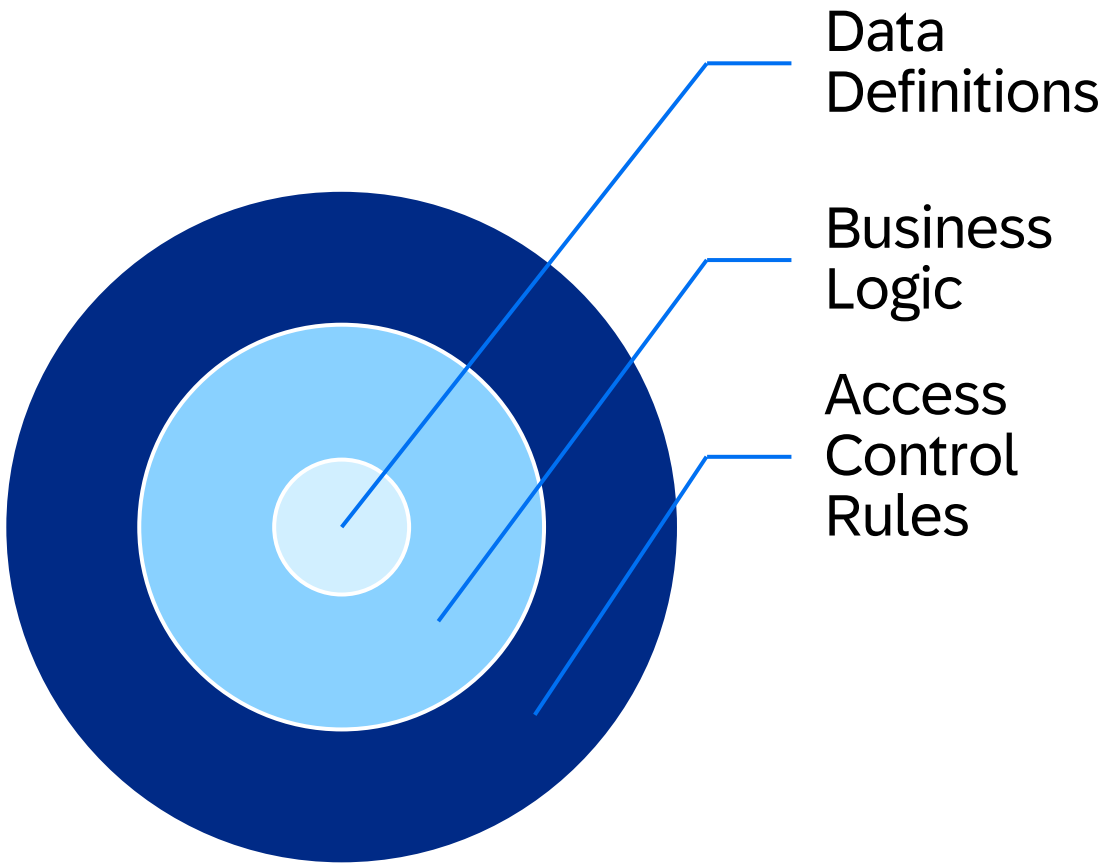
Introduction to Core Data Services (CDS)

By leveraging CDS views, developers can ensure consistency and reusability of data definitions across SAP applications and platforms.

This harmonization of data models not only enhances organizational efficiency but also facilitates easier maintenance and scalability of applications.

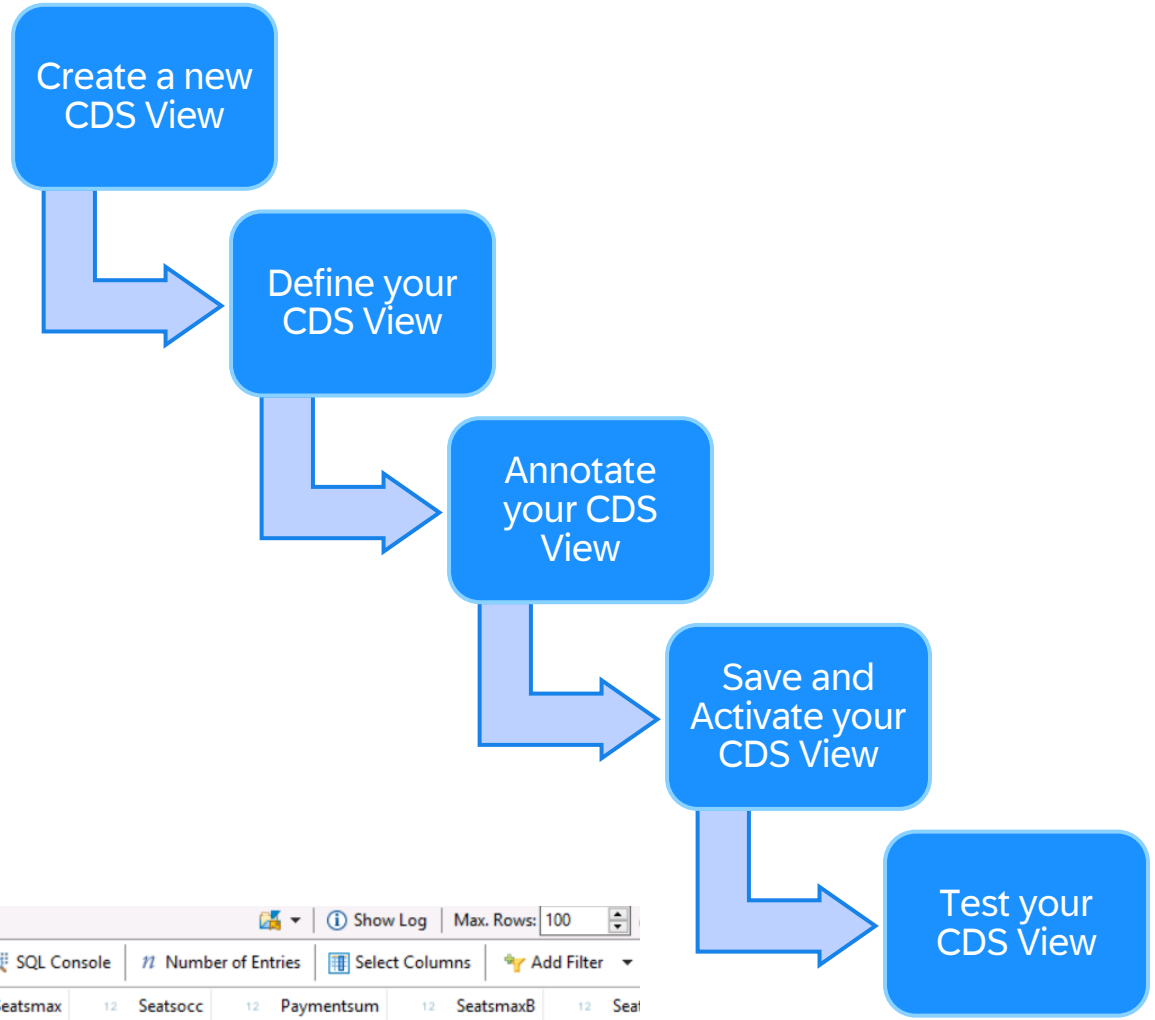
Moreover, CDS plays a pivotal role in OData service generation within SAP environments. By annotating CDS views with OData-specific annotations, developers can automate the creation of OData services.

These services expose CDS views as RESTful endpoints, enabling seamless data integration and consumption across SAP systems and external applications alike.



Steps to Create a CDS View

```
define view entity Z_FLIGHT_CDS as select from sflight
{
    key carrid as Carrid,
    key connid as Connid,
    key fldate as Fldate,
    @Semantics.amount.currencyCode: 'Currency'
    price as Price,
    currency as Currency,
    planetype as Planetype,
    seatsmax as Seatsmax,
    seatsocc as Seatsocc,
    @Semantics.amount.currencyCode: 'Currency'
    paymentsum as Paymentsum,
    seatsmax_b as SeatsmaxB,
    seatsocc_b as SeatsoccB,
    seatsmax_f as SeatsmaxF,
    seatsocc_f as SeatsoccF
}
```



Data Preview | Show Log | Max. Rows: 100

find pattern | 100 rows retrieved - 11 ms (partial result) | SQL Console | Number of Entries | Select Columns | Add Filter

Carrid	Connid	Fldate	Price	Currency	Planetype	Seatsmax	Seatsocc	Paymentsum	SeatsmaxB	Seat
AA	0017	2024-01-04	422.94	USD	747-400	385	367	189976.39	31	
AA	0017	2024-02-05	422.94	USD	747-400	385	365	189443.50	31	
AA	0017	2024-03-08	422.94	USD	747-400	385	374	192746.69	31	
AA	0017	2024-04-09	422.94	USD	747-400	385	367	191714.66	31	
AA	0017	2024-05-11	422.94	USD	747-400	385	368	192433.79	31	
AA	0017	2024-06-12	422.94	USD	747-400	385	365	190615.06	31	
AA	0017	2024-07-14	422.94	USD	747-400	385	373	194510.33	31	



Demonstration 1

CDS View Creation

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OData Service Definitions and Bindings

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Understanding OData V2 Service Definitions

OData V2 service definitions play a crucial role in SAP systems by exposing data models as RESTful services. These definitions serve as a blueprint for how data can be:

- accessed,
- manipulated,
- and presented.

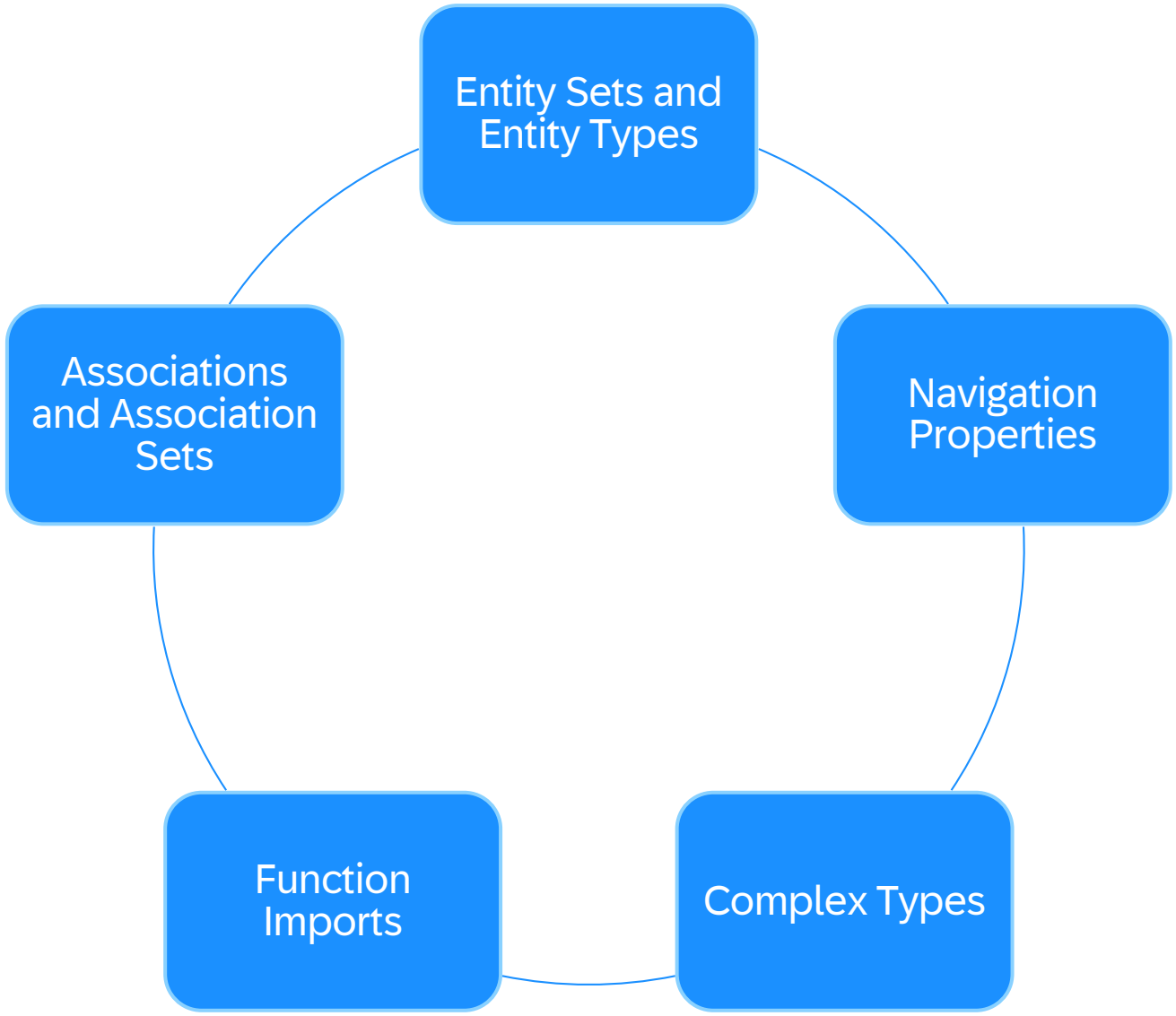
At the core of an OData V2 service definition lies the service metadata document, which delineates the data model and operations available for interaction.

Typically formatted in XML and adhering to OData standards, this document specifies:

- entities,
- relationships,
- and actions supported by the service.

Understanding OData V2 Service Definitions

Key components of the metadata document



Binding OData V2 Services



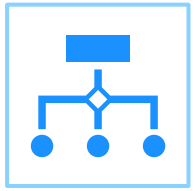
Differences between V2 and V4



Standardization and Compliance

V2: Widely adopted but lacks advanced features; follows initial standards

V4: More mature, aligns with OASIS standards, improves international compliance.



Metadata Handling

V2: Uses XML exclusively, less flexible and verbose

V4: Supports XML and JSON, more efficient and adaptable metadata handling



Data Operations and Query Capabilities

V2: Basic CRUD operations, standard query options; lacks advanced querying

V4: Introduces functions, cross-joins, recursive queries, and aggregation support.

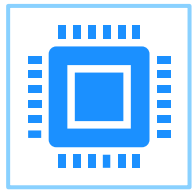


Actions and Functions

V2: Basic support, lacks clear distinction between actions and functions

V4: Clearly distinguishes between actions and functions, enhances expressiveness

Differences between V2 and V4



Batch Processing

V2: Supports batch processing, less efficient

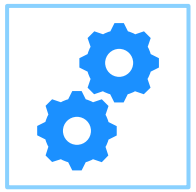
V4: Streamlined mechanism, better support for asynchronous operations



Context and Navigation

V2: Basic context and navigation capabilities

V4: Enhanced support for nested collections and entity relationships



Security and Versioning

V2: Basic security, limited versioning support

V4: Improved security features, robust versioning capabilities

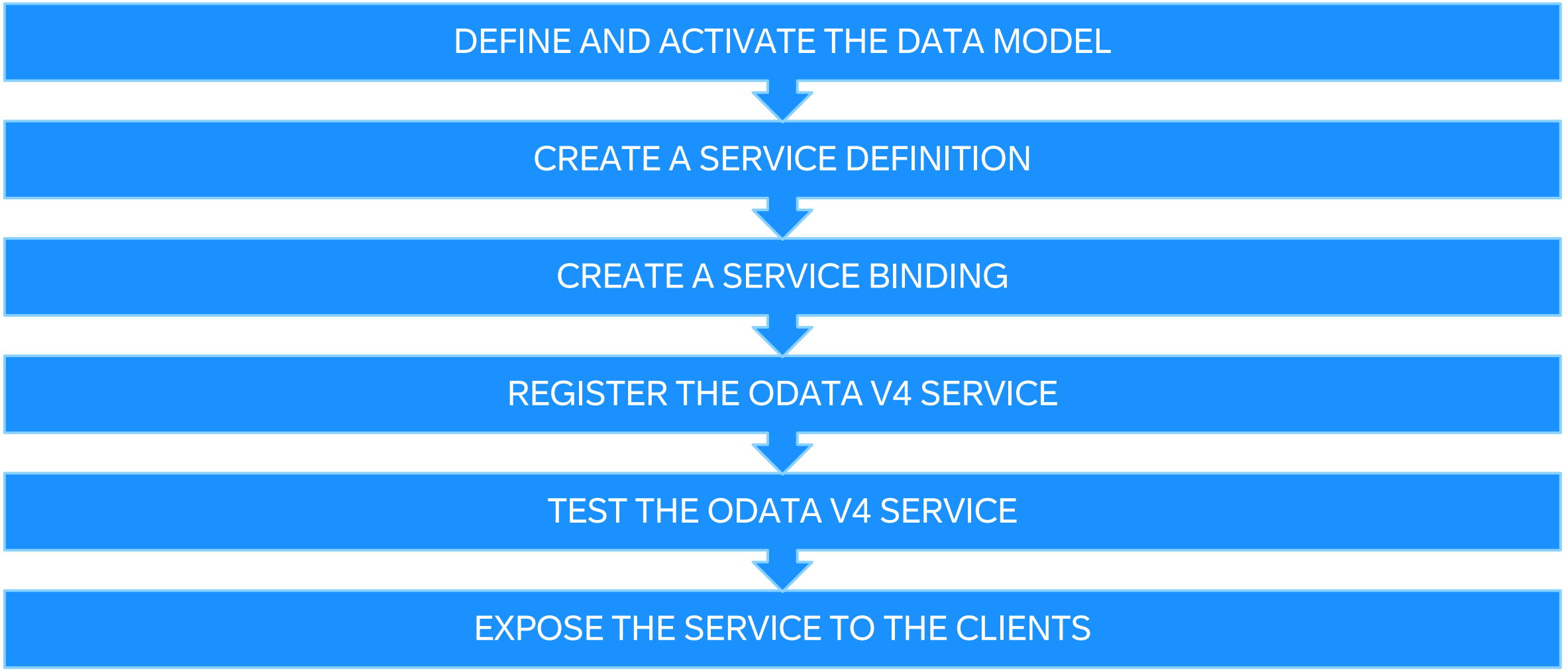


Performance and Scalability

V2: Adequate but can struggle with complex data interactions

V4: Optimizations for improved performance and scalability, supports large datasets

Binding OData V4 Services



RAP Web API binding

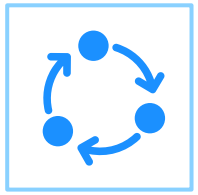
Web API Binding is a type of **OData V4 service binding** specifically designed for **machine-to-machine communication** (APIs), **not for UI**.

- It's used when you want to **expose your RAP object** as a **RESTful API endpoint**
- Ideal for **integration scenarios**, **external systems**, or **system-to-system** data exchange



API Integration

Web APIs in RAP are designed to enable clean, decoupled system-to-system communication.



Metadata

Web APIs focus only on data and operations.



Use Case

Mobile or web apps that need real-time data

Integration with external platforms like CRM/ERP

Middleware tools like SAP Integration Suite or SAP Event Mesh



Extensibility

CDS View extensions for adding fields

BDEF extensions to hook custom logic into save/modify events

Released BAdIs for injecting advanced or partner logic



Demonstration 2

Define and Bind an OData V2/V4 service

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Enhancing OData Services with Annotations

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Overview of OData Annotations

OData annotations enhance the functionality and usability of OData services by providing metadata that describes additional characteristics and behaviours of data elements. This allows for richer interactions between clients and services. Understanding key annotations is essential for creating robust and intuitive data services in SAP environments.

Entity and Property Annotations

@Odata.context

```
<Annotation Term="OData.context"
            String="http://example.com/odata/$metadata#Products"/>
```

@Odata.Community.Display.V1.FormattedValue

```
<Property Name="ProductPrice" Type="Edm.Decimal">
    <Annotation Term="OData.Community.Display.V1.FormattedValue"
                String="$100.00"/>
</Property>
```

Overview of OData Annotations

Capability Annotations

@Capabilities.FilterRestrictions

```
<Property Name="ProductName" Type="Edm.String">
    <Annotation Term="Capabilities.FilterRestrictions">
        <Record>
            <PropertyValue Property="Filterable" Bool="true"/>
        </Record>
    </Annotation>
</Property>
```

@Capabilities.SortRestrictions

```
<Property Name="ProductPrice" Type="Edm.Decimal">
    <Annotation Term="Capabilities.SortRestrictions">
        <Record>
            <PropertyValue Property="Sortable" Bool="true"/>
        </Record>
    </Annotation>
</Property>
```

Validation Annotations

@Validation.Maximum

```
<Property Name="ProductPrice" Type="Edm.Decimal">
    <Annotation Term="Validation.Maximum" Decimal="1000.00"/>
</Property>
```

@Validation.Minimum

```
<Property Name="ProductPrice" Type="Edm.Decimal">
    <Annotation Term="Validation.Minimum" Decimal="0.00"/>
</Property>
```

Overview of OData Annotations

User Interface Annotations

@UI.DisplayName

```
<Property Name="ProductName" Type="Edm.String">
    <Annotation Term="UI.DisplayName" String="Product Name"/>
</Property>
```

@UI.IsReadOnly

```
<Property Name="ProductID" Type="Edm.String">
    <Annotation Term="UI.IsReadOnly" Bool="true"/>
</Property>
```

Semantic Annotations

@Core.IsLanguageDependant

```
<Property Name="ProductDescription" Type="Edm.String">
    <Annotation Term="Core.IsLanguageDependent" Bool="true"/>
</Property>
```

@Core.Description

```
<Property Name="ProductID" Type="Edm.String">
    <Annotation Term="Core.Description"
                String="Unique identifier for the product"/>
</Property>
```

Understanding and utilizing OData annotations can significantly enhance your services' functionality and user experience. They provide valuable metadata guiding clients on interacting with the service, ensuring data is presented meaningfully. Incorporating these annotations will help create more robust, intuitive, and effective data services.

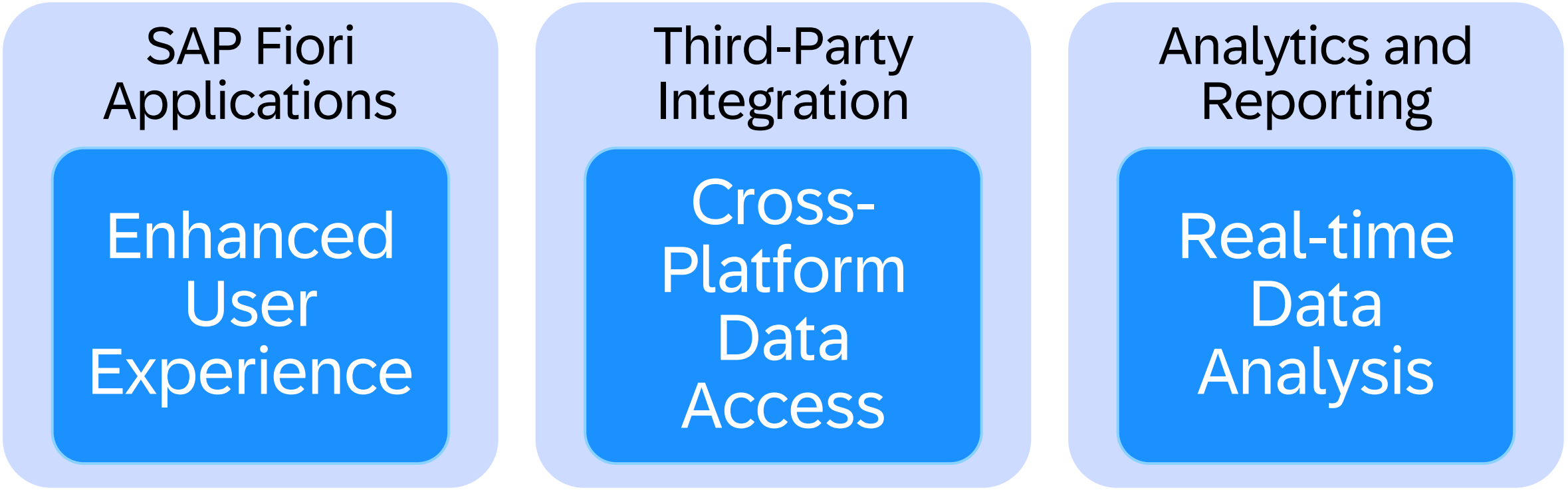
Practical Application and Best Practices

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Real-World Use Cases of OData in SAP

OData has become crucial for data access and integration within SAP, enabling seamless communication between SAP systems and other applications. Real-world use cases include SAP Fiori apps leveraging OData for real-time data retrieval, SAP Gateway facilitating external system integration, and CDS views using OData for exposing complex data models.



Real-World Use Cases of OData in SAP



In summary, OData is vital for efficient data access and integration in SAP, enhancing user experiences, supporting third-party integrations, enabling real-time analytics, and empowering mobile and self-service solutions. Its flexibility and compatibility also make it affective for IoT integrations, demonstrating its versatility in modern SAP implementations.

Best Practices for OData Service Deployment

Deploying OData services in SAP environments requires careful planning and adherence to best practices to ensure reliability, performance, and security. Here are some key best practices for deploying OData services:

1. Designing Efficient OData Services
    - Minimize Data Overhead
    - Implement Paging

```
/Products?$select=ProductID,ProductName,ProductPrice
```

```
/Products?$top=50&$skip=0
```

2. Optimizing Performance
    - Use Batch Processing
    - Leveraging Server-Side Caching

```
<batch>
    <operation method="GET" url="/Products(1)"/>
    <operation method="GET" url="/Products(2)"/>
</batch>
```

3. Ensuring Security
    - Implement Authentication and Authorization
    - Use HTTPS
    - Validate Input Data

Best Practices for OData Service Deployment

4. Error Handling and Logging
    - Provide Clear Error Messages
    - Implement Logging
5. Versioning and Compatibility
    - Support Multiple Versions
    - Deprecate Gradually
6. Monitoring and Maintenance
    - Monitor Performance
    - Schedule Maintenance
7. Documentation and Support
    - Provide Comprehensive Documentation
    - Offer Support Channels

```
{
    "error": {
        "code": "400",
        "message": "Invalid Product ID"
    }
}
```

```
/v1/Products
/v2/Products
```

Resources

OData UI Service Binding based on RAP object

https://help.sap.com/docs/abap-cloud/abap-rap/defining-odata-service?version=sap_btp

Web API binding based on RAP object (can contain custom or released standard objects)

https://help.sap.com/docs/abap-cloud/abap-rap/develop-web-apis

More information on Annotations:
https://sap.github.io/odata-vocabularies/docs/v2-annotations.html

https://cap.cloud.sap/docs/cds/annotations

Resources

SAP Community: ABAP Extensibility Guide – Clean Core (Aug 2025 Update)

https://community.sap.com/t5/technology-blog-posts-by-sap/abap-extensibility-guide-clean-core-for-sap-s-4hana-cloud-august-2025/ba-p/14175399Latest

SAP Community: Clean Core Integration with OData (AIF Monitoring)

https://community.sap.com/t5/technology-blog-posts-by-sap/espresso-bites-clean-core-integration-based-on-an-odata-example-for-sap-aif/ba-p/14287148OData-specific

SAP Help: OData V4 Overview

https://help.sap.com/docs/abap-cloud/abap-rap/odata-v4

Learning Journey: Getting Started with OData V4 RAP Service

https://learning.sap.com/learning-journeys/getting-started-with-creating-an-sap-fiori-elements-app-based-on-an-odata-v4-rap-service/creating-an-odata-v4-service

SAP Help: ABAP RAP Model

https://help.sap.com/docs/abap-cloud/abap-rap/abap-restful-application-programming



Q & A

Thank you.

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